

Main requirement

```
import eel
```

This will probably fail. If so, we may have to install it. In a JupyterLab cell, try:

```
pip install eel
```

And so on for any packages that were not found. Now restart the kernel and try:

```
import eel
```

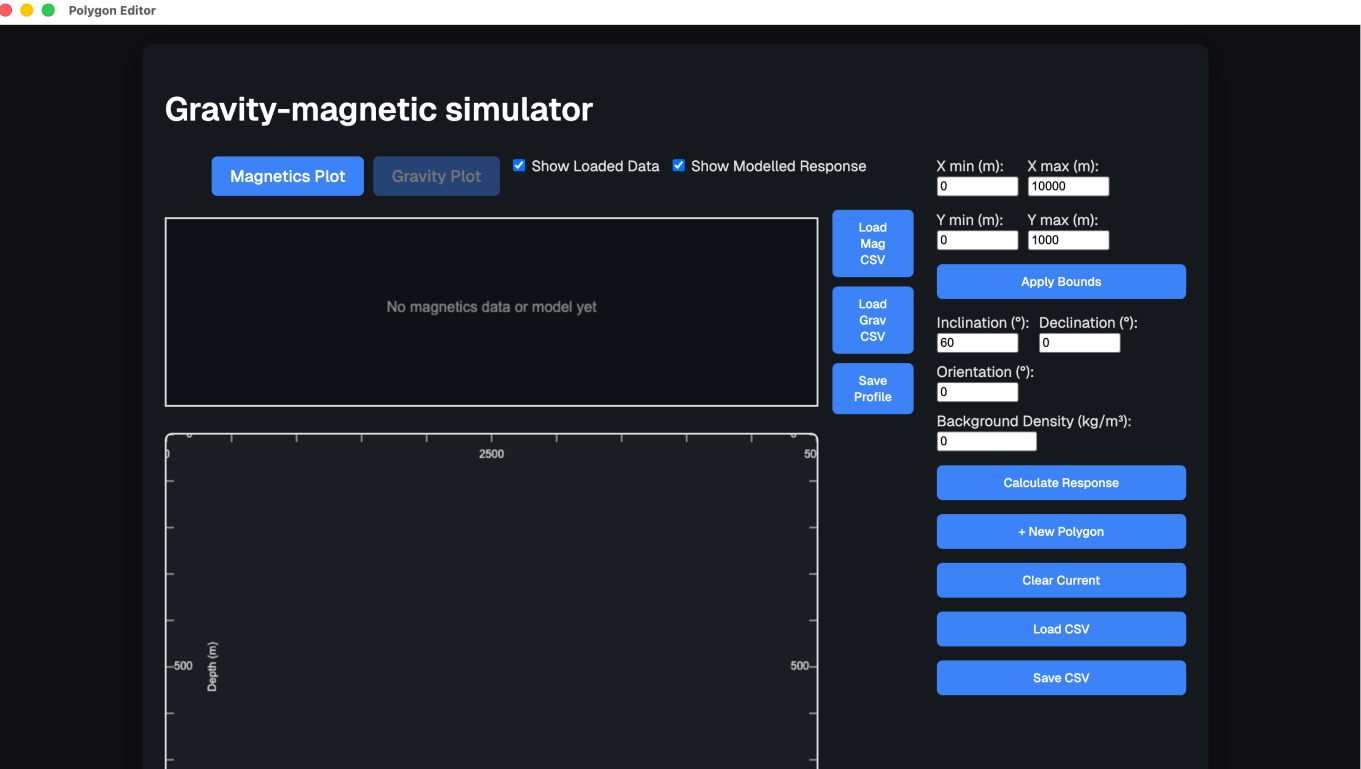
This should work now.

1. Run GravSim2000

Once you have fixed/confirmed this, in Anaconda-> Prompt run:

```
python GravMagSimulator.py
```

And you should get this:



The python script you ran interacts with a html and js file to run on a web browser. The default size is for large scale stuff, let's set it to some appropriate dimensions for our gravity data, and also change the background density:

X min (m):

X max (m):

Y min (m):

Y max (m):

Apply Bounds

Inclination (°):

Declination (°):

Orientation (°):

Background Density (kg/m³):

Now click the button "Apply Bounds". It will rescale your lower canvas area.

2. Model a polygon

Next we're gonna model the magnetic and gravity response of a polygon. The code uses a technique developed by Talwani in the 1960s for this.

> Click "+New Polygon" (red arrow below). You'll see a New Polygon object pop up in the blue box below. The bottom two arrows are pointing to the polygon's properties - magnetic susceptibility (for magnetic modelling), and density (relative to the background) for gravity modelling.

Background Density (kg/m³):

Calculate Response

+ New Polygon

Clear Current

Load CSV

Save CSV

Depth (m)
0
25
50

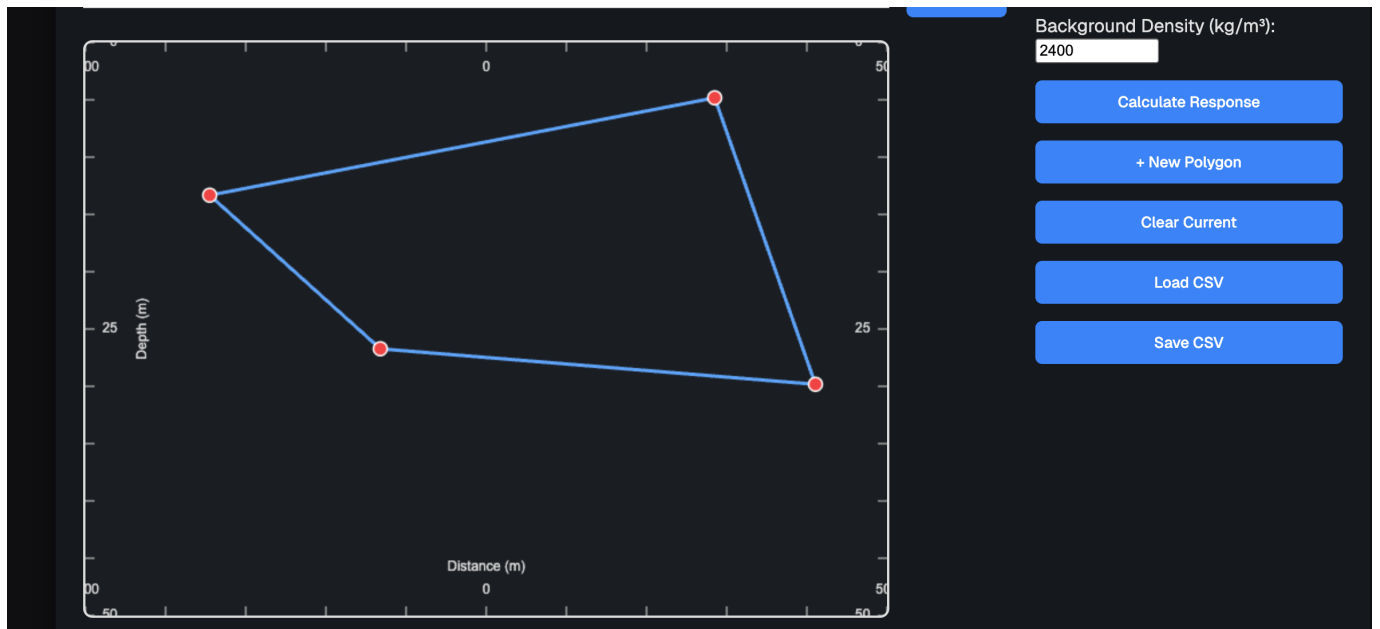
Distance (m)
0
50

Polygons: 1 (0 pts)

Susceptibility (SI) for active polygon:

Density (kg/m³) for active polygon:

Click in the bottom drawing canvas to draw a polygon.



Make sure the points don't cross over or anything weird. If you make a mistake, you can delete the points for the active polygon at the bottom:(click the "x" on the right). Or you can delete the polygon with the "Clear current" button.

1000

Current Points

(-345.000, 13.327)

x

(285.000, 4.827)

x

(410.714, 29.827)

x

(-132.143, 26.727)

x

Once you have a polygon, hit "Calculate response". You'll see a magnetic profile up the top, like this:

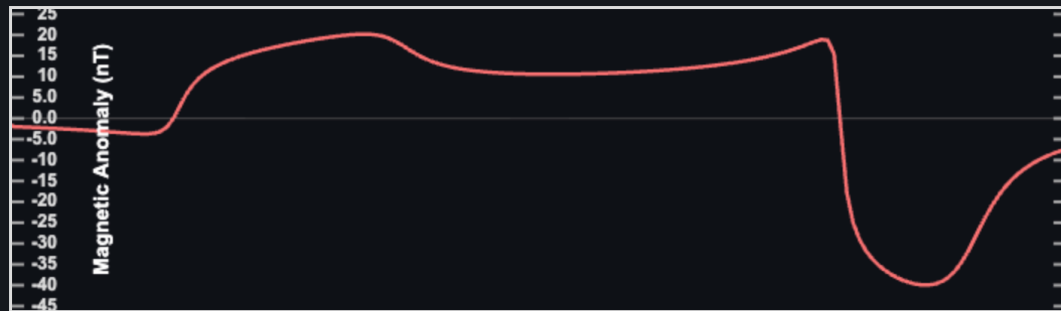
Gravity-magnetic simulator

Magnetics Plot

Gravity Plot

☒ Show Loaded Data

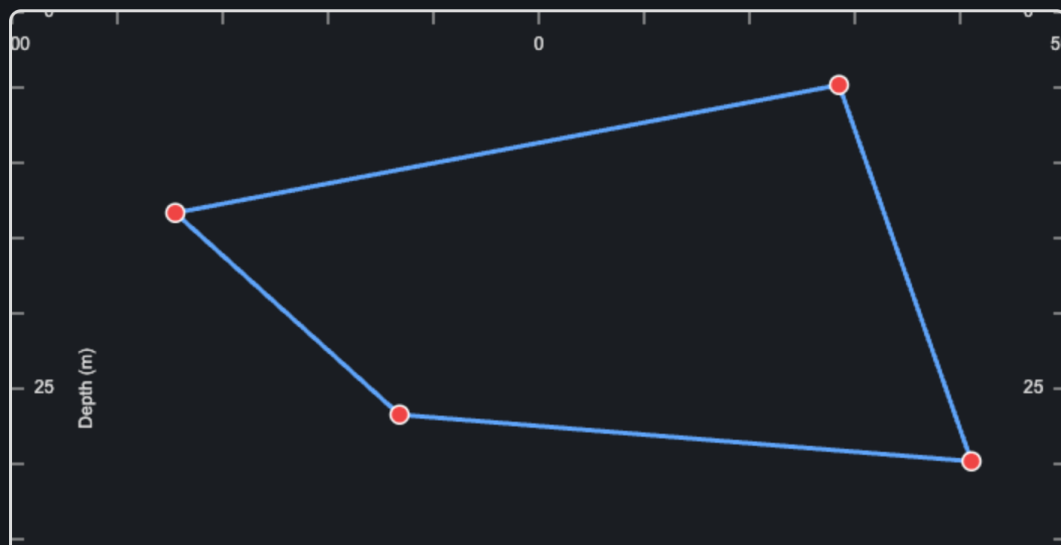
☒ Show Modelled Response



Load
Mag
CSV

Load
Grav
CSV

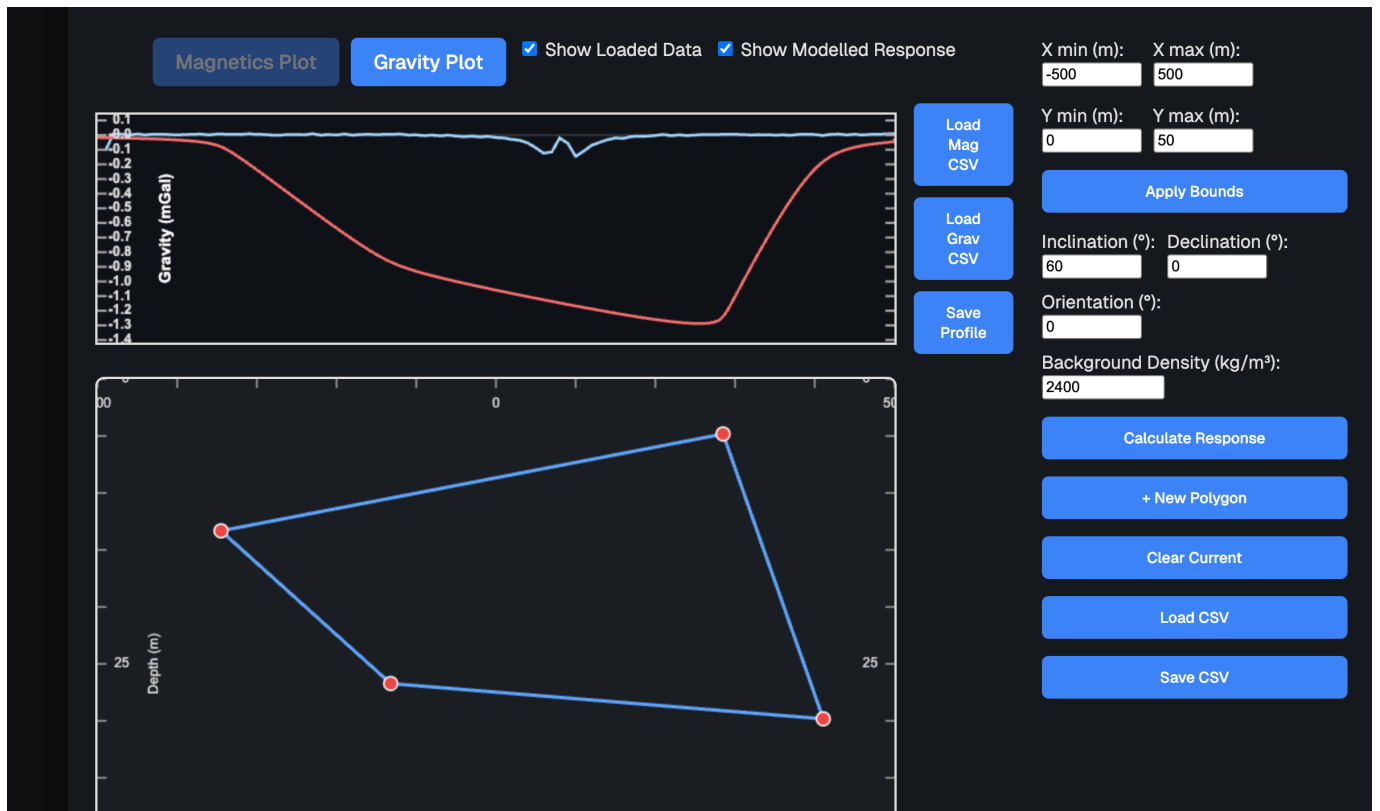
Save
Profile



Click on the gravity plot. You should see the equivalent profile for gravity. Change the density to "1" kg/m³ (appropriate for air). What changes?

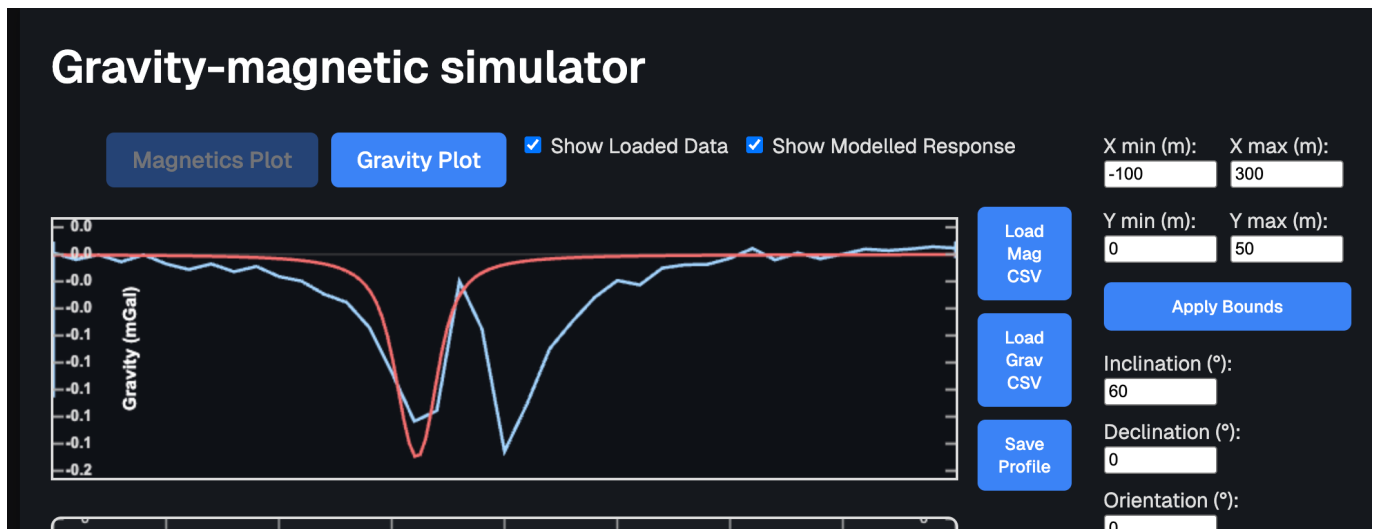
3. Upload your processed gravity data

You should have a gravity profile from the excel spreadsheet, saved as a csv, with two columns - one station (in m), the other gravity (in mGal). Click "Load Grav CSV" to import the data.



4. Model the gravity data

Your task is to work out the shape of the structures giving the gravity anomaly in the profile you processed. Draw polygons, change the density, and calculate the gravity response, to try and fit the profile shape. What do you think these structures are?



Hints and Tips:

1. The software is beta. It may hang occasionally. If it does, force quit and start again. It seems more stable if you import the csv first, then draw polygons. I need to debug why!
2. Try varying your X min and X max to fit the anomaly in better (see above). Don't forget to click "Apply Bounds".
3. Density can't go to zero, but it can go to 1kg/m³ - which is effectively air in the calculation.
4. "Save profile" will save your modelled profile as a csv for later plotting, etc.

5. To submit

The results of this exercise will form part of the assignment 1 submission. Make sure you have

- 1) A full plot of your processed gravity data
- 2) A side-by-side comparison of your field gravity data, and your modelled gravity data (this can be in jupyter notebooks, or screenshot

from the Simulator screen).

3) A picture of your polygons showing the modelled response fit on top.

4) An interpretation of the structure/polygons that give an optimal fit for this anomaly.